Retail Sales Performance Analysis

Superstore Dataset — End-to-End Analytics Project

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1. Executive Summary

This report analyzes 9,993 retail transactions (2014–2017) across four U.S. regions to identify revenue drivers, profitability gaps, and loss-making patterns. The dataset was cleaned and validated in Excel, analyzed using SQL queries in PostgreSQL, and visualized through an interactive Power BI dashboard.

Total Sales: $2,296,957 | Total Profit: $286,324 | Total Orders: 5,008 | Overall Profit Margin: 12.5%

2. Regional Performance

The West region leads in both sales and profitability, while the Central region shows a concerning gap between revenue and margin.

Region	Total Sales	Total Profit	Profit Margin
West	$725,215	$108,346	14.9%
East	$678,781	$91,523	13.5%
Central	$501,240	$39,706	7.9%
South	$391,722	$46,749	11.9%

3. Category Profitability

Technology delivers the strongest margins. Furniture stands out as a high-revenue, low-margin category that merits a pricing or discount policy review.

Category	Total Sales	Total Profit	Profit Margin
Technology	$836,154	$145,455	17.4%
Office Supplies	$718,804	$122,418	17.0%
Furniture	$741,999	$18,451	2.5%

4. Customer Segment Contribution

- Consumer segment generates 50.6% of total revenue ($1,161,158) — the primary growth driver.
- Corporate segment contributes 30.7% ($706,146) with healthy profit conversion.
- Home Office is the smallest segment (18.7%) but maintains a comparable margin to Corporate.

5. Seasonal Sales Trend (2014–2017)

- Sales grew consistently year-over-year: $484K (2014) to $733K (2017).
- November and December consistently post the highest monthly sales — peak: November 2017 at $118,448.
- January and February are consistently the weakest months across all four years, indicating a post-holiday slowdown.

6. Loss-Making States — Discount Risk

Nine states recorded a net loss over the analysis period. In every case, the average discount rate exceeds 28%, indicating discounting is eroding margin rather than driving profitable volume.

State	Avg. Discount	Total Loss
Texas	37.0%	-$25,730
Ohio	32.5%	-$16,971
Pennsylvania	32.9%	-$15,560
Illinois	39.0%	-$12,608
North Carolina	28.4%	-$7,491

7. Recommendations

- Review discount policy in Texas, Ohio, Pennsylvania, and Illinois — discounts above 30% are consistently loss-making in these markets.
- Investigate Furniture cost structure or pricing — high sales volume is not converting into proportional profit.
- Increase marketing spend in Sep–Nov to capitalize on the proven seasonal demand spike ahead of the holiday period.
- Use the Central region's lower margin (7.9% vs. West's 14.9%) as a case study to identify operational or logistics inefficiencies.

8. Methodology

- Data Cleaning: Removed duplicates, standardized date formats, validated discount and profit ranges in Excel (9,993 of 9,994 rows retained after cleaning).
- Database: Imported cleaned data into PostgreSQL; wrote SQL aggregation queries to validate regional, category, segment, and state-level KPIs.
- Visualization: Built a 6-visual interactive Power BI dashboard with a year slicer for drill-down analysis, cross-validated against Excel pivot tables.